

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Forming & Solving Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FORMING & SOLVING EQUATIONS · CH2 · L13

ENG. ESLAM AHMED



The Map

This lesson collapses Forming & Solving Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 7 website questions, 7 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x + 6 = 30 - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1HR Q6

Use trigonometry

$$4x + 6 = 30 - x$$

Finish: $x = 4.8$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 6 2 10 p (c) Find the value of p p = (1) 5 1 r = (d) Write down the value of r r = (1) 6
The diagram shows triangle ABC A B C Diagram NOT accurately drawn (4x + 6) cm
(30 - x) cm CA x 4 6 cm CB x 30 cm angle angle CAB CBA = Work out the value of x x
=

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4(5 - x)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P2H Q2

Area of rectangle A

$$4(5 - x)$$

Area of rectangle B

$$5(2x - 1)$$

Finish: $x = 2.5$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

2 Here are two rectangles, rectangle A and rectangle B. Diagram NOT accurately drawn

4 cm	5 cm
(5 - x) cm	(2x - 1) cm

rectangle B rectangle A

The area of rectangle B is twice the area of rectangle A. Work out the value of x. Show your working clearly.

$$X =$$

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x + 6 = 30 - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1HR Q6

Use trigonometry

$$4x + 6 = 30 - x$$

Finish: $x = 4.8$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

6 6 2 10 p (c) Find the value of p p = (1) 5 1 r = (d) Write down the value of r r = (1) 6
The diagram shows triangle ABC A B C Diagram NOT accurately drawn (4x + 6) cm
(30 - x) cm CA x 4 6 cm CB x 30 cm angle angle CAB CBA = Work out the value of x x
=

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.